1 Limitations

The axiom view, and SAI as its formal model, leave out much of what scientific theories are and do. For each limitation we say whether it is partially addressable, and by what, or a genuine limit of a declarative view. The judgements are arguments, not theorems; sentences that rest on a theorem cite it.

1.1 Statistical confrontation with data

In physics a theory meets data statistically: with models of the instruments and the noise it assigns likelihoods to data, often near-raw (**??**), and statistical theories answer with probabilities, tested against frequencies. The proof-requiring core of SAI has no native noise model: a datum is proved or not. Graded fit enters only through randomised theories in M_A, such as laws with calibrated error models (**????**); through the generative variant, whose updating we prove given a support condition and whose identification we only sketch (**????**); or through sentences about probabilities, whose relation to finite frequencies is not deductive. It matters for our computations: on noise-free interval data the syntax of the law texts decides among the forms compatible with the Mars data, and with graded fit the data decide (**????**). So the proof-requiring core is the noise-free special case; its natural extension inside M_A is a law with a calibrated error model; a native treatment is open.

1.2 Written theories and their readers

A written theory joined to a simple reasoner does not solve the problems the theory is used for. We think this is true, and see three gaps. *The bridge gap:* a problem posed in ordinary terms (a rope over a pulley) becomes formal only through idealisations and bridge principles that the theory does not contain (cf. Cartwright, 1983; Morgan and Morrison, 1999); **??** prices the bridge from raw data to questions as the complexity $K(\eta)$ of an extractor, large in real cases. *The computational gap:* for the formalised problem, theory and mathematics suffice logically, but finding the answer needs methods, lemmas and approximation schemes, as in deriving "Kepler's laws hold within ε on a window" from Newton's theory (**??**). In the model of **??** this gap is a theorem, visible only when what is written is what is charged: program length sees neither reasoner nor vocabulary (**??**), while charged as written the size of a usable theory depends exponentially on the pair (reasoner, vocabulary) (**??**); charged deduction, which bounds a derivation that exists without charging its search, cannot express the gap (**??**, with the derivation sizes sketched in **??**). So insufficiency for a simple reasoner is a property of the pair (theory, reasoner) and of the vocabulary, when what is written is what is charged. This is not a theorem about textbooks (**??**); it shows that a declarative account of a weak reasoner's competence must include the lemmas, not that people carry exponentially many of them. *The tacit gap:* perceptual and manual skills (Polanyi, 1958, 1966) and exemplars, through which new problems are seen as like solved ones (Kuhn, 1970, 1977), are not sentences; in SAI exemplars are only data. This is a genuine limit.

1.3 Approximate truth

Kepler's laws as stated are false. The difficulty this seems to pose for a proof-requiring inducer is a limitation of proof-requiring compatibility with deterministic theories, which rewards the false exact law over its true weakening (**??**), not of the axiom view: inside M_A an approximately true law is a law with a calibrated error model (**??**). Whether a deterministic theory holds a law

honestly or with exceptions depends on how exceptions and silence are priced, a modelling choice (**??**), and humility is partiality for deterministic theories, with the trichotomy as its formal home, and calibration for randomised ones (**??**). Open are a canonical metric for grading approximate predictions, which we think does not exist, since purposes set the tolerances, and a general comparative notion of truthlikeness, which is known to be hard: on Popper's definition no false theory is closer to the truth than another (cf. Popper, 1963; Miller, 1974; Tichý, 1974), and the similarity approach needs a metric (cf. Niiniluoto, 1987). Our notion is deliberately local: a question family, a tolerance, a domain.

1.4 Domains, several models, and inconsistent theories in use

Practice uses mutually inconsistent models of one system, each in its domain (the Earth as flat, as a sphere, as an oblate spheroid), while proof-requiring SAI gives a globally inconsistent theory weight 0 (classical electrodynamics of point charges has been argued to be inconsistent, Frisch, 2004, which is disputed). A sound device exists: state each model in its honest form, "q lies within ε of f on the domain", and reason classically. Over a true background that decides the comparisons involved, exact models that give one length the values 100 and 99.9 prove everything, while honest forms of true models are jointly consistent and give true conclusions with propagated tolerances (**????**). But the device neither explains practice nor reproduces it: physicists rarely state or propagate tolerances, worst-case propagation becomes vacuous over long chains and for chaotic dynamics, and the rigorous honest form of the Boltzmann equation, Lanford's theorem as described in expositions (Lanford, 1975), holds only for a fraction of the mean free time in the low-density limit, while practice uses the equation far beyond that, and rightly. The device blocks the catastrophe, but does not say how to choose a model, a domain or a tolerance, and would forbid much correct practice; we take that gap to be part of what written theories leave to tacit understanding. Under SAI's pricing a domain usually appears as a tolerance with an error distribution that varies with the system, not as a sharp boundary (**????**). Knowing when a model applies is a recognizer, a genuine limit.

1.5 Concepts

Theories are built from concepts. Table 1 lists features of human concepts that an account of concepts as pieces of code, combined to generate predictions, misses, and says whether the axiom view captures them.[1] We accept them all as features of human concepts, with two qualifications: that inferential relations are constitutive is one major account (cf. Sellars, 1953), disputed by referential accounts (cf. Kripke, 1980) and supported for theoretical terms by the Ramsey–Lewis treatment (Lewis, 1970); and natural types we find plausible. One feature supports the axiom view: if a notion can have different recognizers, its identity is fixed otherwise, by inferential role or reference.

For analogy the universal prior gives a bound. Let $K(T)$ be the least $K(e)$ with $\mathsf{W}_e = T$. For every r.e. theory T_1 and computable map τ from sentences to sentences,

$$K\big(T_1 \cup \tau[T_1]\big) \;\leq\; K(T_1) + K(\tau) + c, \tag{1}$$

since an index of $T_1 \cup \tau[T_1]$ is computable from indices of T_1 and τ, and $K(x,y) \leq K(x)+K(y)+c$: a theory with its image under an analogy costs at most $K(\tau)+c$ bits more than the theory alone. Structure mapping (Gentner, 1983) and the models of science (cf. Hesse, 1963) are accounts of

[1] The list follows, with changes, an unpublished draft essay of Hänni's on concepts.

feature of concepts	axiom view	how captured, or what is missing
for questions and problems, not predictions	yes/no	yes for questions posed in a language (**??**); no for problems needing search or action
inert alone; parts of a mind	a limit	worth depends on reasoner and vocabulary (**??**); free deduction hides it
inferential role constitutive	yes	axioms are inferential relations (meaning postulates, Ramsey–Lewis)
compositional	yes/no	yes for contents, as formulas; no for recognizers, which are procedures
learning by new links	partly	a persisting symbol, but a global mixture over whole theories
soft; meaning change	partly	"within ε of a sphere" is a true partial theory; not meaning change
analogy	partly	partial interpretations, rewarded as they compress (1); not finding them
new concepts from old	partly	definitions: $O(1)$ universally (**??**), useful to a chaining reasoner (**??**)
methods attached	partly	lemmas for a bounded reasoner, or axioms via Craig at exponential cost (**??**); not motor methods
recognizers	no	a procedure on situations; the extractor of **??**, taken as given
one notion, many recognizers	yes	identity is the symbol with its axioms; recognizers are bridges
built from pointers	yes	a constant fixed by one axiom, which gathers further axioms
purposes	no	axioms can state functions; their use in search is not modelled
clarification; facing action	no	not in the inducer
natural types	partly	sorts and arities; their emergence is not modelled

Table 1: Features of human concepts, and whether the axiom view captures them (an assessment).

such partial interpretations. A name whose reference is fixed by a description, like Le Verrier's "Neptune" (cf. Kripke, 1980), is a pointer. Concepts that behave in patches (cf. Wilson, 2006) or stretch under counterexamples (cf. Lakatos, 1976), and interpretation as abduction (Hobbs et al., 1993), are outside the view.

1.6 Other limitations

Science predicts very particular things, in set-ups it builds. SAI captures half: its data are selected sentences and its theories partial. Which questions are worth asking, and the making of phenomena by experiment (cf. Hacking, 1983), are outside it; questions weighted by value would be a handle on the first. *Concept formation, discovery and understanding.* Step 3 of the ladder (**??**) is partially addressable (**????**); step 4, the revision of the base abstractions, is a genuine limit for now. SAI is an ideal of the evaluation of theories over a fixed space of hypotheses, not of discovery or understanding: in **??** the pool of forms is given, not found (on discovery systems, see **??**). *Theory-laden and fallible data.* A base language may presuppose a theory (heliocentric longitudes presuppose Copernican geometry; **??**), which biases comparisons between theories that disagree about it. And SAI takes labels as correct: a deterministic theory must prove even a mislabelled datum, by an exception axiom; a noise component in the likelihood, as in Hänni and Claude (Anthropic) (2026), is a partial remedy. *Purposes and technology* (engines, circuits, lenses) are outside the view, a genuine limit. *Science as a social structure,* in which theories are communicated, divided, checked piece by piece and extended, is a genuine limit of a model with one inducer, and may be where the question of **??** finds its answer. *The mathematics is free, and large* (in **??**, ordered fields with trigonometry). Building a background B' into B gains at most $K(B') + c$ bits on any data (**??**), so charging for it changes little in the universal setting; we expect it to matter much more under charged deduction. And with free deduction, data that the background entails carry no information: if $\mathsf{Q} \subseteq \mathrm{Cn}(B)$, checking instances $P(\underline{0}), \ldots, P(\underline{N})$ of a Δ_0 property leaves $\mathrm{Bel}(\forall x\, P(x))$ at its prior (**??**), the problem of old evidence and logical omniscience (cf. Hacking, 1967; Garber, 1983). Induction in mathematics needs a reasoner for

whom deduction is costly.

What the axiom view captures. It captures the static, declarative and inferential side of theories and concepts: contents, inferential relations, the composition of contents, learning by weighing axioms (globally, not concept by concept), pointers, partiality, and approximate truth in honest or error-model form. It misses the procedural side (recognizers, methods, the search for derivations, action), the statistical side of the contact with data in its proof-requiring core, and the dynamic side (purposes, softness, clarification, meaning change, discovery, the social process). That is less than a theory of concepts asks for, but we do not offer one. We offer a model of the declarative core of scientific theories, in which the procedural side reappears formally as the dependence of a written theory's worth on its reasoner and its vocabulary (**??**).

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